

Obstruction Certificates as Dual Objects: One Template, Four Instantiations

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

September 23, 2026

Abstract

The obstruction calculus issues four families of non-viability certificates — common-action, timing, fibre-certification, and certainty-equivalence — that were derived separately. This paper, the D6 item of the generalization catalogue, unifies their computational reading: each certificate is a *dual feasibility object* for the primal system the corresponding obstruction declares infeasible. The common-action certificate is a Farkas multiplier vector; the timing certificate is a telescoping (additive) or substitution (multiplicative) identity on the decline rows; the fibre certificate is a contradicting-witness pair inside one observation cell; the certainty-equivalence certificate is an affine drift lower bound. On the programme’s working instances all four dual objects are computed and validated in exact rational arithmetic, together with the template’s scope statement: duals certify nonviability only, viability being certified primitively by witness actions — and at the corner state where no witness exists, primal and dual agree in declaring nonviability (6/6 exact check families).

Introduction and relation to the programme

The completed items derive their certificates mechanism by mechanism. The source catalogue proposes the applied-mathematics reading adopted here: *the obstruction calculus produces primal-dual certificates of nonviability under information constraints*. The paper makes that reading exact on the programme’s own audited instances — no new mechanism is introduced, and no mechanism from the completed items is modified; the contribution is the identification, the exact witnesses, and the scope statement.

The template

Each obstruction family declares a primal feasibility problem infeasible:

- **Common action:** the stacked safe-action system $Au \leq b$ (one instrument must serve several states’ floors);
- **Timing:** the decline-chain program $z_t \geq 1$, $z_{t+1} = z_t - \varepsilon$ (or its multiplicative contraction $q_{t+1} = \frac{9}{10}q_t$);

- **Fibre certification:** a single exact label for all states of one observation cell;
- **Certainty equivalence:** a law in the CE class whose corrected drift stays below the viability threshold.

The template: an obstruction certificate *is* a dual object — a multiplier vector, a telescoping identity, a witness pair, an affine lower bound — each checkable in exact rational arithmetic against its own primal.

Proposition 1 (Common action). $Au \leq b$ is infeasible iff a rational $\lambda \geq 0$ exists with $\lambda^\top A = 0$, $\lambda^\top b < 0$; the multiplier vector is the obstruction certificate (Farkas’ lemma).

Proposition 2 (Timing). For the additive chain, summing the K decline rows with all-ones multipliers yields the identity $z_0 - z_K = K\varepsilon$, whence survival for K steps holds iff $z_0 \geq 1 + K\varepsilon$; for the contraction chain, substitution yields $q_K = q_0(\frac{9}{10})^K$, whence survival iff $q_0 \geq (\frac{10}{9})^K$ — the level sets of the successor paper’s exit-time value. The dual objects are the identity rows themselves.

Proof. Telescoping sums of equalities are equalities; the multiple case is the same statement with multiplication. Both directions are exact, so the threshold identities hold with equality at the boundary. $\square$

Proposition 3 (Fibre and CE). If one observation cell carries two states whose exact labels differ, the cell admits no exact certifier, and the dual object is the contradicting pair. If every law in a policy class has drift bounded below by an affine function exceeding the viability threshold on the audited grid, the class is empty as a correspondence with safety, and the affine bound is the dual object.

Proof. Immediate from the definitions; the content is the exact instances below. $\square$

Verified instances (6/6 exact check families)

E1. For the two-floor stacked system $u \leq \frac{2}{5} \wedge u \geq \frac{3}{5}$: $\lambda = (1, 1)$, $\lambda^\top A = 0$, $\lambda^\top b = -\frac{1}{5} < 0$.

E2. Additive thresholds $z_0 \geq 1 + K/10$ ($K \leq 4$) and multiplicative thresholds $q_0 \geq (\frac{10}{9})^K$ ($K \leq 3$) verified at exact powers and $\frac{1}{100}$ -neighbors — equality at the boundary in both cases, matching Proposition 2.

E3. The crossing fibre of the two-floor index instance carries $s = \frac{1}{10}$ (label low) and $s = \frac{2}{5}$ (label high) in one cell $\mathcal{O} = 1$: no exact certifier exists; the dual object is the pair.

E4. On the CE trap with reading $g(s) = s^2$ and correction $\frac{1}{10}$: the drift is exactly $g(s + \frac{1}{10}) - g(s) = \frac{s}{5} + \frac{1}{100}$ on the 101-point grid, and $\frac{s}{5} + \frac{1}{100} \geq \frac{21}{100}$ iff $s \geq 1$ — the affine dual bound is exact.

E5 (template closure). Every dual object re-validates its primal infeasibility by direct evaluation; the feasible contrast system ($u \leq \frac{2}{5} \wedge u \geq \frac{1}{5}$) admits the witness $u = \frac{1}{5}$ and no certificate: primal and dual agree on both sides of every verdict.

E6 (scope). On the two-floor two-patch system, safe states off the corner $(1, 1)$ have primal witness actions; at the corner both primal (one-step exit under every instrument) and dual (its stacked system is trivially infeasible) certify nonviability. Duals certify nonviability only; viability is never dual-certified.

What is not claimed

The template is an identification, not a new algorithm: no solver is developed, no complexity claim is made, and the barrier-style/SOS dual objects mentioned in the source catalogue for nonlinear systems are cited as outlook, not computed (the programme’s instances are discrete and rational). Sum-of-squares methods, semidefinite duals, and continuous-state duality are out of scope. The four instantiations share a lemma (Farkas) only at the level of template; no deeper converse is claimed — in particular the paper does not claim that every nonviability proof in the programme must dualize.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `certificate_duality_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.